

JYOTHSNA LI-TECH PVT LTD

Application-Optimized LFP Cells for India's Energy Infrastructure

Built for high heat, long life and reliable solar, UPS, telecom and commercial backup applications.

Up to 55C

Target operating environment

4,000-6,000 cycles

Lifecycle target

INR 8-12 Cr

Initial raise

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ENERGY

India: Jyothsna Li-Tech Pvt Ltd, Bangalore

China: Dongguan Jialite Trading Co., Ltd.

Registered office: No. 60, 12th Cross, Agrahara Dasarahalli, Magadi Main Road, Bangalore 560079



India's backup infrastructure has a reliability gap

Lead-acid systems and generic lithium packs struggle under high heat, frequent cycling and downtime-sensitive use.

- 40-55C operating environments reduce battery life and system reliability
- Lead-acid replacement cycles every 18-24 months create recurring cost
- Maintenance visits, downtime and warranty failures hurt EPC economics
- Import-linked pricing and supply risk make long-term planning difficult

Customer pain

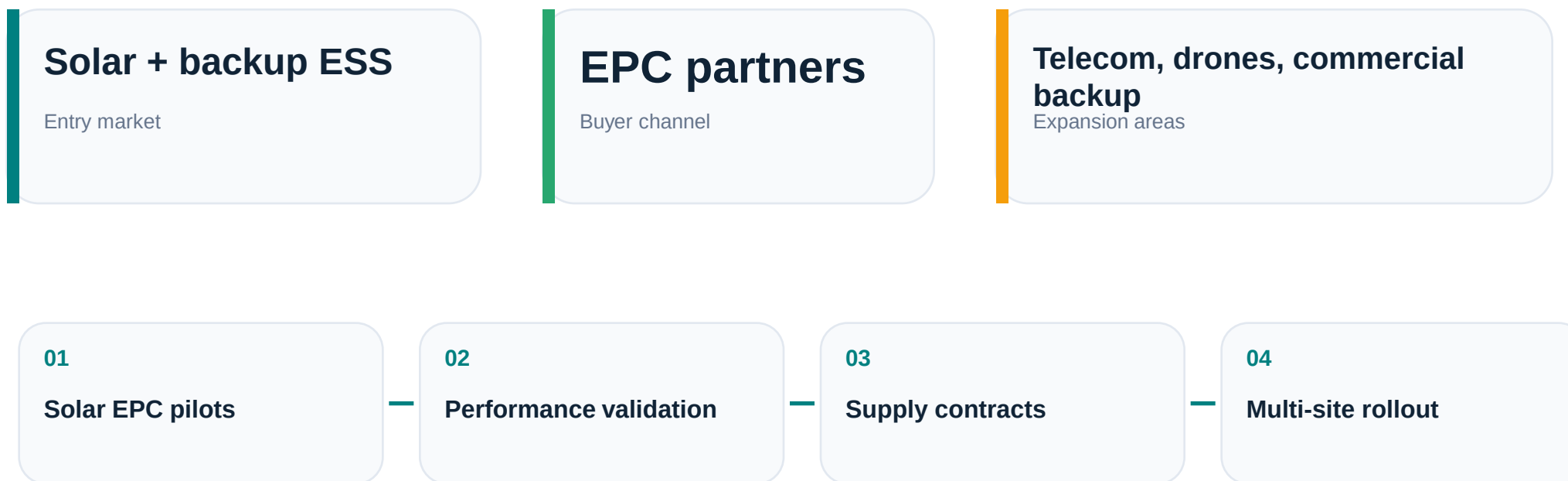
The purchase price is not the real cost.

Replacement, service, heat failure and downtime decide lifetime economics.

JLT focuses on lifecycle cost, not commodity battery pricing.

Start narrow, expand wide

The first wedge is B2B ESS and backup systems sold through solar EPC and infrastructure partners.



The go-to-market avoids expensive direct consumer selling and uses EPC partners that already own installation, service and customer relationships.

Independent Customer Validation

Verace Market Research: buyer interviews and purchase intent signals.

160

Customer interviews

97%

Purchase intent

77%

Willing to pay premium

Key customer pain points

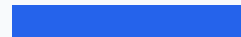
Lead Time



MOQ Restrictions



Import Dependence



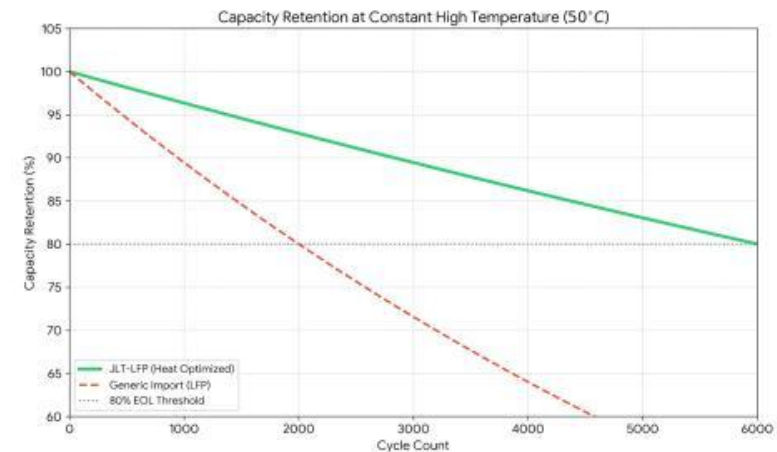
Cost Volatility



JLT-LFP: purpose-built cells for infrastructure duty cycles

Application-specific LFP cell design for ESS, UPS and solar storage in harsh Indian conditions.

- Optimized for high-temperature operation and long cycle life
- Designed around ESS and backup duty cycles, not automotive use cases
- Supports pack makers and EPC partners with localized engineering
- Aims to reduce total cost of ownership through fewer replacements



Capacity retention: JLT-LFP vs Lead Acid

Product specification targets

LFP prismatic cells positioned for lifecycle, safety and stability.



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Prismatic LFP Cell Platform

50-200Ah

Capacity range

140-160 Wh/kg

Energy density target

4,000-6,000

Cycle life target

Up to 55C

Operating target

Target applications

- Solar ESS
- UPS and commercial backup
- Telecom backup
- Battery pack OEMs

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Lower lifetime cost through fewer replacements

The case for JLT is won on lifecycle economics, maintenance reduction and downtime protection.

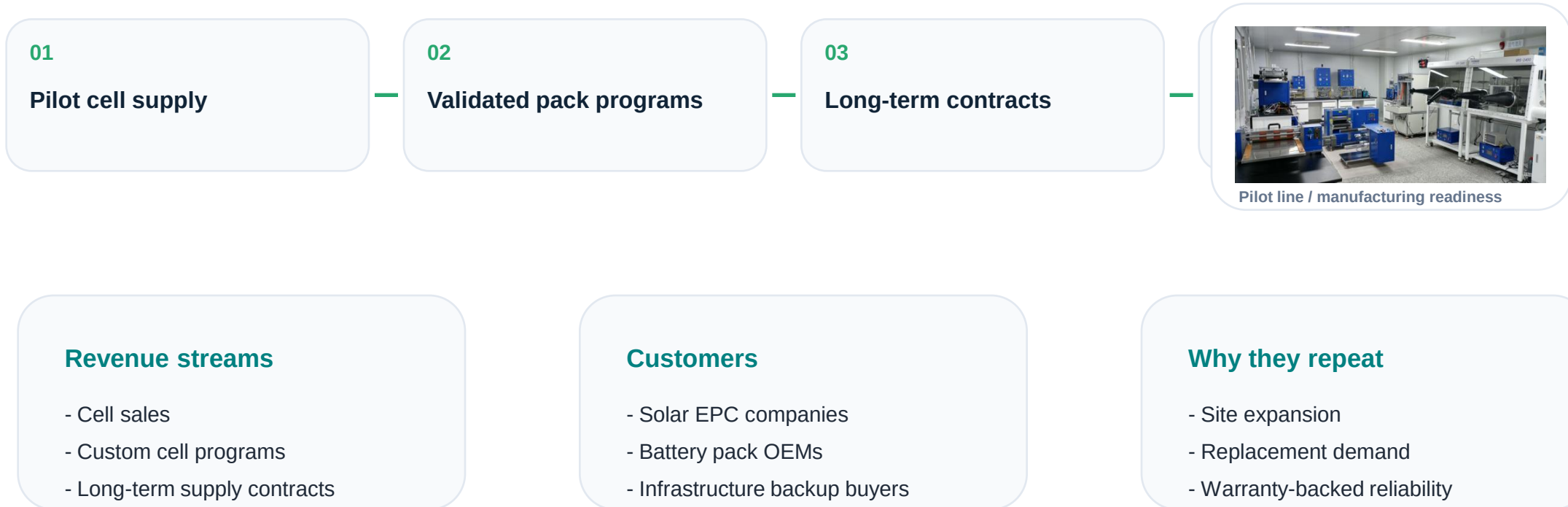


System	Initial	Replacement	Maintenance	Downtime	5-year total
Lead-acid	INR 80k	INR 160k	INR 25k	INR 20k	INR 285k
Cheap lithium	INR 120k	INR 60k	INR 10k	INR 10k	INR 200k
JLT-LFP	INR 140k	INR 0	INR 5k	INR 3k	INR 148k

Assumptions should be shown in appendix or investor Q&A: system size, usage profile, temperature, warranty and service model.

Cell sales plus long-term infrastructure supply relationships

Recurring demand is driven by replacement cycles, site expansion and OEM/pack-maker partnerships.



Focused B2B entry through EPC partners

Use pilot installations to prove thermal reliability, lifecycle value and service economics.



- Initial wedge: solar, UPS and commercial backup systems
- Partner-led distribution reduces customer acquisition cost
- Field data becomes a sales asset for future deployments
- Localized support improves confidence versus import-only suppliers

Control without unnecessary technology risk

JLT combines access to proven manufacturing know-how with in-house product optimization.

- Exclusive access to proven lithium-ion manufacturing know-how
- In-house application engineering for Indian ESS duty cycles
- Global raw-material sourcing strategy for cost efficiency
- No dependency on external technology ownership for core execution

More field deployments create better thermal performance learning and customer trust.

Control stack

01

Design

02

Process

03

Testing



Manufacturing know-how access

Dongguan Jialite Trading Co., Ltd. supports China manufacturing ecosystem access.

Execution readiness is moving from setup to market entry

Evidence-backed execution status and post-pilot customer path.

DPIIT recognized

Karnataka project support

Land allocation secured

Machinery quotation available

Pilot funding active

Customer qualification post-pilot

- Technical team aligned for execution
- Customer discussions initiated
- Pilot deployment conversations underway
- LOI engagements in progress

Government Recognition & Readiness

Institutional readiness signals for investor confidence.

DPIIT Startup Recognition

Recognized

Startup India recognition strengthens institutional credibility.

Karnataka Project Support

Approved

State-level support for lithium-ion cell manufacturing project.

Land Allocation

Secured

Land allocation referenced for project execution readiness.

Execution Readiness

Active

Machinery quotations, pilot funding and customer qualification pathway.

Manufacturing Setup

Planned

Pilot line, dry room, testing and working cap funding ask.

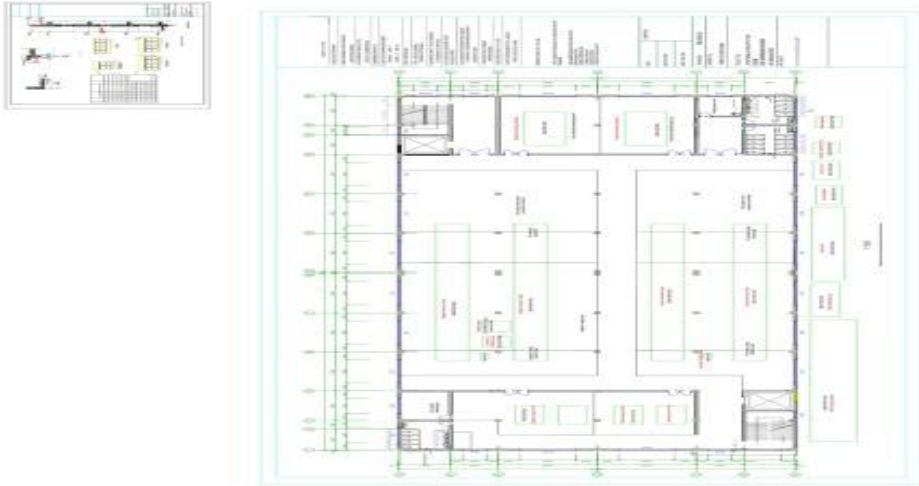


Credibility is built around approvals, land readiness and execution milestones, not early-stage claims.

Land & Location Readiness

Project location and land readiness evidence from actual project documents.

Project layout / site plan



Land status

- KIADB to allot 1 acre of land
- Sira Industrial Area
- Tumkur District
- Government order references infrastructure support



Fig 1: Map indicating the study area

Focused where large players are broad

JLT targets underserved infrastructure ESS segments where customization and local support matter.

JLT

ESS-optimized cells
Designed for 45-55C
4,000-6,000 cycles
High customization

Imported LFP Packs

Generic mass-produced
Heat degradation risk
2,000-3,500 cycles
Import dependent

Indian Pack Integrators

Cells outsourced
Limited cell control
Cell-dependent life
Medium customization

Lead-Acid

Legacy chemistry
Severe heat degradation
500-800 cycles
High maintenance

Positioning: purpose-built infrastructure cells for pack makers and EPC-led deployments in Indian operating conditions.

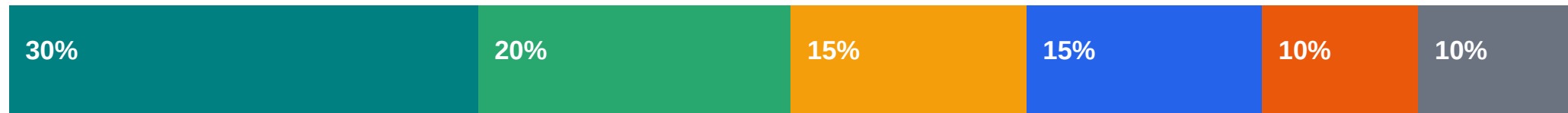
Capital-efficient scale strategy

Revenue growth by phase, with capacity expansion.



Raising INR 8-12 Crore

Use of funds allocation totals 100%.



 Pilot Equipment - 30%

 Dry Room - 20%

 Testing & Certification - 15%

 Raw Materials - 15%

 Technical Team - 10%

 Working Capital - 10%

- Pilot line setup
- Testing and certification
- Technical team expansion
- Working capital for initial customer programs

Founder-led execution with cross-border manufacturing access

Founder credibility, China ecosystem access and execution readiness.



Narendra Srirams

Founder & Managing Director

- 24+ years industrial manufacturing experience
- China manufacturing ecosystem since 2006
- Technology access secured in 2018
- Founder investment exceeding INR 20 Cr